



UNIVERSITÀ
DI TRENTO



Master's Degree in Cognitive Science

NEURAL MASS AND SPIKING MODELS FOR THE STUDY
OF NEURAL CIRCUITS

Supervisor:
Albrecht Haase

Student:
Zeno Rossi

Academic Year 2025/2026

Contents

- 1 Stochastic Resonance in a Spiking Neural Network Model of the Honeybee Antennal Lobe** **1**
 - 1.1 Introduction 1
 - 1.1.1 Classical Stochastic Resonance 1

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Chapter 1

Stochastic Resonance in a Spiking Neural Network Model of the Honeybee Antennal Lobe

1.1 Introduction

One of the first well established findings in neuroscience with the advent of electrical recordings of neuronal activity, was the variability of neurons' responses. That is, when stimulated with the same input, neurons respond with different temporal patterns of spikes, while the overall firing rate remains roughly equal [1], [2], [3].

This trial-to-trial variability is commonly called noise. Sources of noise in neurons are many and not all variability in response is due to noise (see [4] for a review). Still, due to the physical scale of cells, a stochastic component in neuronal function is inevitable by thermal noise alone. It is then to be expected that neural systems would have evolved to tolerate or, more daringly, exploit the presence of noise.

1.1.1 Classical Stochastic Resonance

The idea of studying the effects of noise on the dynamics of a system first appeared in the work of Benzi et al. [5] and Benzi et al. [6], where the term Stochastic Resonance (SR) was used to explain the periodic behavior of earth's average temperature. A brief covering of that first SR application will be made, as it provides a practical understanding of the general phenomenon and a surprising link to neural systems.

The most common way to model temperature averages was using simple energy balance deterministic models:

$$C \frac{dT}{dt} = R_{in}(T) - R_{out}(T) \quad (1.1)$$

where C is the earth's thermal capacity, and R_{in} and R_{out} represent the total incoming and outgoing solar radiation, defined as

$$R_{in} = Q\mu \quad (1.2)$$

$$R_{out} = \alpha(T)Q\mu + \varepsilon(T) \quad (1.3)$$

where μ is the average incoming solar radiation.

In [6] a simplification is used: 3 steady-states are assumed to exist as $T_1 < T_2 < T_3$, those are possible earth's climates. 1.1 then becomes

$$C \frac{dT}{dt} = \frac{\mu \varepsilon(T)}{1 + \beta(1 - T/T_1)(1 - T/T_2)(1 - T/T_3)} - \varepsilon(T) \quad (1.4)$$

To analyze the stability of the defined climates the rate of change of temperatures is taken as

$$F(T) = \frac{dT}{dt}, \quad (1.5)$$

then a pseudo-potential energy function is defined as

$$\Phi(T) = - \int F(T) dT \quad (1.6)$$

that represents the "energy" needed to change the temperature across temperature values. From Φ we get

$$F(T) = - \frac{\partial \Phi}{\partial T} \quad (1.7)$$

Then intuitively the maximum of Φ will be an unstable point, while the minima will be stable. That is, where the second derivative with respect to temperature

$$\frac{dF}{dT} = - \frac{\partial^2 \Phi}{\partial T^2} \quad (1.8)$$

is < 0 , any neighboring starting state of the earth's temperature will converge to it, while when it is > 0 it will diverge.

It can be shown that T_1, T_3 are stable points, while T_2 is unstable: a peak any starting state would "roll off" from (see fig). In our context, T_1, T_3 are the only observable climates, temperatures at which the earth will stay for enough time to be detectable in the geological records. This type of system where 2 stable states exist is called a bistable system, and the plot of Φ over T displays the typical double well shape of the (pseudo)potential function.

With the present model, we can see that the earth's climate would depend solely (all else being equal) on the starting temperature, i.e., whether it is smaller or greater than T_2 . Then it would follow $F(T)$ towards T_1 or T_3 . However, earth's temperatures do indeed show oscillations with a period of 10^5 years.

A weak cosinusoidal forcing modulating the input solar radiations can be introduced, representing the varying distance of the earth from the sun along its elliptic orbit. Then μ becomes time dependent

$$\mu(t) = 1 + \gamma \cos(\omega t) \quad (1.9)$$

where $\gamma \ll 1$ and $\omega = 2\pi/10^5$, reflecting the observed oscillation period. The derivative and pseudo-energy function become $\tilde{F} = \tilde{F}(T; t)$, $\tilde{\Phi} = \tilde{\Phi}(T; t)$.

Still, this periodicity alone does not lead to any change in climates: the system never jumps over T_2 (see fig).

Here the turning point, if gaussian white noise, representing various mechanisms causing T fluctuations at a smaller time scale, is added to the system, yielding

$$\frac{dT}{dt} = F(T) + \sigma \eta(t) \quad (1.10)$$

then for certain values of σ a jump between stable states will become possible, with a frequency $\approx 2\pi/10^5$.

Bibliography

- [1] G. J. Tomko and D. R. Crapper, "Neuronal variability: Non-stationary responses to identical visual stimuli," *Brain Research*, vol. 79, no. 3, pp. 405–418, Oct. 25, 1974, issn: 0006-8993. DOI: 10.1016/0006-8993(74)90438-7. Accessed: Jun. 14, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0006899374904387>.
- [2] Z. F. Mainen and T. J. Sejnowski, "Reliability of Spike Timing in Neocortical Neurons," *Science*, vol. 268, no. 5216, pp. 1503–1506, Jun. 9, 1995, issn: 0036-8075, 1095-9203. DOI: 10.1126/science.7770778. Accessed: Jun. 14, 2026. [Online]. Available: <https://www.science.org/doi/10.1126/science.7770778>.
- [3] D. J. Tolhurst, J. A. Movshon, and A. F. Dean, "The statistical reliability of signals in single neurons in cat and monkey visual cortex," *Vision Research*, vol. 23, no. 8, pp. 775–785, Jan. 1, 1983, issn: 0042-6989. DOI: 10.1016/0042-6989(83)90200-6. Accessed: Jun. 14, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0042698983902006>.
- [4] A. A. Faisal, L. P. J. Selen, and D. M. Wolpert, "Noise in the nervous system," *Nature Reviews Neuroscience*, vol. 9, no. 4, pp. 292–303, Apr. 2008, issn: 1471-0048. DOI: 10.1038/nrn2258. Accessed: Nov. 27, 2025. [Online]. Available: <https://www.nature.com/articles/nrn2258>.
- [5] R. Benzi, A. Sutera, and A. Vulpiani, "The mechanism of stochastic resonance," *Journal of Physics A: Mathematical and General*, vol. 14, no. 11, p. L453, Nov. 1981, issn: 0305-4470. DOI: 10.1088/0305-4470/14/11/006. Accessed: Mar. 29, 2025. [Online]. Available: <https://dx.doi.org/10.1088/0305-4470/14/11/006>.
- [6] R. Benzi, G. Parisi, A. Sutera, and A. Vulpiani, "Stochastic resonance in climatic change," *Tellus*, vol. 34, no. 1, pp. 10–16, 1982, issn: 2153-3490. DOI: 10.1111/j.2153-3490.1982.tb01787.x. Accessed: Jun. 16, 2026. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.2153-3490.1982.tb01787.x>.